Building Information Modeling (BIM) Workflows for Construction Sequencing & 4D-Planning of 3D-Printed ISRU Surface Habitats (Abstract 715110)

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ABSTRACT

Future autonomously constructed surface habitats on the Moon and Mars will leverage the versatility of additive manufacturing technologies to create airtight and durable structures which protect the crew from the harsh extraterrestrial environment. BIM workflows present significant value in the design and engineering of many space systems but particularly so for habitat facilities—which will include pre-integrated hardware components such as ECLS, airlocks, suitports, as well as hygiene modules. Synergies between BIM and construction management softwares with robotic control and software systems for emplacement and for additive manufacturing are yet to be realized—however there is no question that this research will become a necessary milestone in the development of autonomously constructed habitat systems. In particular, 4D-simulation in BIM software lends tremendous value to mission planning and mission architectures which conceive the autonomous robotic construction of a habitat system. In 2019 team SEArch+/Apis Cor won first place in Virtual Construction (BIM) Level 2 (100% Design) within NASA's Phase 3 Challenge for a 3D-Printed Habitat on Mars, which was judged on criteria such as: BIM use functionality, Completeness of 3D model according to distinct levels of development for retaining walls, MEP, and ECLSS, Realism of 4D Model, among others. This paper will define the landscape of BIM use and simulation as it will apply to the future robotic deployment of space habitats, followed by a description of the project's contribution to BIM workflows for the design of an ISRU surface

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habitat in the short term (both for developing an appropriate mission architecture as well as for full-scale construction prototyping on Earth), and concluding with recommendations for next steps in advancing present research and construction prototyping with such softwares.

INTRODUCTION

Future long duration exploration missions will benefit from robotic precursor missions which autonomously construct surface infrastructure and habitat systems prior to a crew's arrival. For autonomously constructed habitats to successfully deploy on the Moon and Mars, the construction sequence must consider: required site conditions relevant to the foundation system, quantities of raw materials required at the site for ISRU collection, feedstock storage and access to the mixing apparatus for 3D-printing, access to power sources, as well as many other environmental and in-situ based resources necessary for autonomous deployment and construction of the habitat. In SEArch+/Apis Cor's submissions to NASA's Phase 3 Centennial Challenge for a 3D Printed Habitat on Mars, the team merged Building Information Modeling (BIM) workflows between Rhino/Grasshopper, Autodesk Revit, Navisworks, and a proprietary slicer software developed to estimate time to completion for additive construction as well as material quantities for feedstock.

Uses of BIM in AEC. BIM is an essential component within the digital management of construction projects. The advantages of BIM are that it enables a holistic collection of linked building information, provision of data, and integration of building models drawn from shared databases containing detailed information of the building at every phase of its lifecycle. Central storage and planning with BIM can be considered state-of-the-art, and software programs enable integration of data sources and types with the building model itself. A major shift in philosophy in working with BIM tools as opposed to other digital computer aided design (CAD) tools is the conceptualization of every drawn element as not representative lines and geometry but rather as a building element in itself. This fundamental re-conceptualization of design as building object elements rather than representations has made it possible to embed metadata into a project previously impossible in 2D and 3D CAD drawing. BIM information ranges from proximity and location of objects to material characterization and/or properties, quantity takeoff, assembly and sub-assembly information, followed by time for construction. It provides a seamless platform for real-time collaboration and sharing of information embedded within geometric metadata associated with 3-dimensional building components. Within the Architecture, Engineering, and Construction (AEC) industries BIM is used extensively in project planning and execution for applications such as clash detection of virtual geometries (Teizer, 2018). However, the uses of BIM span from early project planning and design, to operational facilities management and lifecycle analysis for completed projects.

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Interoperability is a pillar of BIM because the information or metadata contained within models needs to be exchanged (usually across different software) in order to be useful. IFC (Industry Foundation Classes), a computer developing standard, is often cited as the most common means by which BIM interoperability can be achieved (Santos, 2010). A model can be converted into IFC data schema, which is a text based structure representing all the data nested in a model such as geometry, material type, existing family parameters (e.g. robots) and hierarchy of elements along with project metadata (Davtalab, 2018). Despite the strengths of IFC and its prominence within a number of design-build projects and examples within the field, interoperability and the integration of BIM platforms with automated construction systems remains an ongoing challenge.

Challenges of BIM-Integrated Frameworks for Automated Construction. While BIM has been largely adopted on large-scale and complex building projects, and is generally responsible for the development and integration of new computational technologies and software within AEC, it has not followed with direct opportunities for automation and/or the use of robotics in construction (Correa, 2016). Implementation in full-scale automated building construction is still necessary. BIM alone is not sufficient to accomplish directed planning of automated construction processes on and off-site (Meschini, 2018). BIM within AEC has been traditionally conceived as a tool for project planning and scheduling rather than an integrated approach or framework for the coordination of automated construction processes—suggesting that there is a lack of specific software to adequately represent the complexity of modern buildings, and to plan and test automated construction processes (Meschini, 2018). Information may certainly be shared and transferred between BIM models and numerical control outputs, but the data acquisition, data types, and organization are yet to be streamlined or addressed comprehensively for automated and/or autonomous construction. This is in part, of course, because robotic systems and platforms tend to be devised for a particular application or hardware setup, in which, more often than not, software is designed specifically for the robotic control system. Automation is appealing for construction because it can guarantee work quality, reduce the cognitive load and burden on project management, achieve greater labor efficiency, and ensure productivity without fatigue. Automation in construction will also become more widely adopted as a necessity to meet the growing demand for buildings while the supply of skilled labor continues to decline.

3D vs 4D BIM. BIM software frameworks provide a means of handling construction schedules both in the planning and execution phases (Tauscher, 2014). 3D BIM objects, containing valuable metadata, may be correlated to a construction schedule in a fairly automatic and generative way, to produce a 4D construction visualization. 4D planning and sequencing includes the simulation and control of detailed construction site logistics processes, such as site layout planning, the inclusion and removal of temporary equipment or materials, and more ("3D-Printed Habitat…Competition Rules"). 4D BIM schedules assign construction tasks to building (geometry)

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elements, part of the underlying BIM, and derive a task order or sequence for the construction tasks within a schedule.

The application of BIM within planning stages has been demonstrated and verified within AEC industries as a means of simulating logistics and process workflows within construction. In actual construction stages and progress monitoring during building construction, many gaps remain in developing real-time updates to BIM to capture deviations and anomalies that inevitably occur within construction processes. One of BIM's ongoing shortcoming is that of construction progress detection (Asadi, 2018). Errors and changes to design plans within as-built structures are rarely captured within record models. Tools for efficient (automated) data collection on construction sites and (near) real-time analysis are still missing (Teizer et al, 2017). After completion of the construction phase, BIM is important as an integrated representation for lifecycle analysis within operations and management.

BIM for Manufacturing. BIM enables software to convert virtual 3-dimensional geometry into Computerized Numerical Control (CNC) codes which dictate the 3-dimensional positioning of a 3D-print head, robot arm, end effector, or any type of additive deposition mechanism, generally speaking. In this way, BIM-based software enables CAD to CAM production. The methodology of using BIM for fabrication enables the assembly motion plans for CNC in robotic assembly and prototyping. G-code is the most widely used and well-known numerical control programming language for 3D-printing, used for many automation processes and manufacturing tasks involving machine tools. G-code defines the speed, trajectory of motion (tool path) and the selection of the specific tool. BIM methods extend traditional CAD-based 3D modeling workflows, by seamlessly integrating metadata associated with 3D geometries, in addition to other processes for planning, scheduling, operations and maintenance (Teizer, 2018). While BIM could certainly function to link data structures between CAD / CAM / CAE in mechanical engineering, it has not functioned as such in professional practice to date and the examples of such applications are very few (Correaa, 2016).

3D-Printing and BIM. Additive manufacturing has the potential to disrupt the status quo for automation in construction today. Workflows from BIM to automated additive construction will augment productivity marking a giant leap in the efficiencies provided to automation within AEC. 3D printing is a disruptive technology set to upturn construction practices, but many have yet to develop a strong business case to be able to implement the technology. Given the risk-adverse nature of the construction industry, in order for 3D-printing to become a field-ready construction method, computer control of the manufacturing sequence will best integrate with logistics for other aspects of a project including the integration of MEP elements, materials schedules, and more.

BIM software, as developed for the AEC industry, has been organized to represent largely planar geometries for wall, floor, and roof families functioning as

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multi-material sub-assemblies in construction management. While BIM tools in the AEC industry can represent simplified geometries for additive manufacturing, the representation of material extrusion through machine- and tool-paths would serve to further integrate BIM tools and additive manufacturing construction methods. This re-conceptualization would further make it possible to integrate a 4D BIM workflow visualizing the continuous extrusion of material.

BIM for Habitat Design. In a majority of circumstances, integration from design to production between BIM and robotics is achieved with the use of IFC Schema for the BIM model with building design, and the representation of the printing process for that design, and geometry within a STL file to be used with a 3D Printer (Correaa, 2016). Additionally, the 3D-printer itself will more likely than not be only one of multiple robotically deployed elements at the construction site within a future mission scenario. Site logistics including layout planning using required facilities for automated construction will likewise be necessary to consider and integrate within a unified control system. The integration of CAD and BIM softwares with robotic tools such as Robot Operating System (ROS) could indeed represent the next step in realizing automated construction processes in the AEC industries (Meschini et al, 2016). ROS is an open-source operating system for robots originally developed in 2007 by the Stanford Artificial Intelligence Laboratory.

System Architecture Logistics for Surface Site Infrastructure on Mars. Before 3D-printing is deployed for infrastructure development, site establishments will rely on construction machinery for excavation, levelling, grading, and preparation of the terrain. Space robotics will be an integral part of surface site preparation relevant to ISRU for the construction of deep space infrastructure. Robots will assist in scouting and surveying build and excavation sites, in addition to prospecting and processing raw materials such as regolith. Tasks such as material transport, equipment positioning and assembly will be managed by robot fleets on precursor missions (Fong et al, 2005). Machinery for site preparation will not only need to mobilize autonomously, but connect and integrate with other ISRU systems (such as for power).

All the same, the design and planning of 4D construction sequencing for earth-based construction and in deployment within a future Mars surface mission requires precise knowledge from human operators and planners of available workspaces for automated machines and robots, and requires knowledge of intelligent safety solutions (such as a collision-free work environment based on real time location sensing) for appropriate and intelligent human-machine interaction to occur (Teizer, 2018). Autonomous operations do not obviate systems engineers and designers from having expert knowledge of additive manufacturing standards ensuring structural integrity at the building scale, as an example, and the present need to have construction crews monitoring construction automation activities (particularly so in 3D-printing) in real-time points to the great challenge of an autonomously deployed robotic Mars precursor mission. Ensuring resilience and autonomy is a goal for many

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deployable mission architectures for future long-duration exploration missions. Artificial intelligence combined with BIM will provide a useful reference for decision-making as the degree of autonomy develops (Kang, 2018). Fault detection and quality assurance in 3D-printing will likewise benefit from machine learning and computer vision for recognition.

Construction Progress Through BIM. Because construction sites tend to be highly dynamic and uncontrolled environments, achieving or simulating automation and autonomous operations on-site and without human intervention has been a distinct challenge. Robots operating autonomously in unstructured environments will require precise methods to locate building components on-site, and object detection mechanisms are to-date not well suited to construction applications (Dawod, 2019). The challenges that come with adopting robotics within building construction highlights why industrial applications in closed environments with repeated tasks have been better-suited to robotic applications. BIM models not only contain databases relevant to 3-dimensional object recognition for vision systems, but also provides the physical and mechanical specification about the object and their spatial relation with other elements in the design, which helps in robot guidance and pose estimation.

Realizing Autonomous Construction through BIM. Vision capabilities and perception systems are essential for robotics to adapt to continuously changing site conditions and enable accurate object recognition and estimation on construction sites (Dawod, 2019). Efforts in computer vision have focused on extracting geometry and element information from static BIM models (Ferguson, 2019). Moreover, creating and maintaining a dynamically updated model of the building environment remains a difficult challenge (Ferguson, 2019). The full extent to which BIM models can function as a "digital twin" of as-built or in-progress construction sites and buildings to automatically update a large-scale geometric BIM model in real-time will continue to be examined in current research. Automated data capture methods are likely to improve surveying systems in the AEC industries. Advancements in computer vision and 3D object recognition will certainly bridge gaps in information and data transfer for BIM models and how they can how they capture or represent activity at constructions sites in real-time. BIM databases will thus be able to represent not only design configurations and schedules, but also deviations and changes.

NASA's Centennial Challenge for a 3D-Printed Habitat on Mars. The 3D-Printed Habitat Challenge is part of NASA's Centennial Challenges Program and focuses on both habitat design as well as technology development of large-scale additive construction systems capable of fabricating structures from in situ materials (such as regolith) and/or mission recyclables (such as plastic packaging) (Prater et al, 2018). The Centennial Challenge program issued solicitations in 2015 and 2018 for virtual designs relevant to building information modeling (BIM) for additive construction. The Phase 2 Challenge asked teams to develop material mixtures and 3D-printing systems and produce test specimens (compression, flexure) and manufacture a small

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dome structure (Prater et al, 2018). The Phase 3 3D-Printed Habitat Competition was subdivided into multiple Virtual Design as well as Construction submission levels, with the ultimate intent of advancing the applicability of BIM to large-scale additive manufacturing projects in space. SEArch+ / Apis Cor's participation within NASA's Phase 3 3D-Printed Habitat Challenge enabled the team to contribute and advance this area of research.

Virtual Construction Levels of NASA's 3D-Printed Habitat Challenge. In the Virtual Construction levels of NASA's competition, teams were asked to create BIM models of autonomously constructed habitat structures and provide detailed information on materials, design, and construction sequencing (Prater et al, 2018). Mars X-House by team SEArch+ / Apis Cor won first place within Final Virtual Design (100% Design) within the Phase 3 NASA 3D-Printed Habitat Challenge (see Figure 3 below). Mars X-House celebrates innovation in radiation shielding techniques while allowing natural light to penetrate the structure, supporting the astronauts' physiological and psychological well-being in a long-duration mission (Yashar et al, 2019). Our human-centered approach prioritizes safety, redundancy, and the wellbeing of the crew. Considerations such as tensile loading, printability as well as simulation of an appropriate construction sequence given a mobile printing apparatus factored into the design decisions made for the habitat.

Figure 1. Mars X-House Exterior View.

Requirements. The habitat designs required a pressure-retaining living space of at least 93 m^2 with the intent of supporting four astronauts for one year. Teams were asked to deliver a BIM model as well as a 4D construction sequence simulating all autonomous activities at the site including: construction machinery, additive manufacturing, and emplacement of pre-integrated components (ECLS, airlocks, etc). Teams were asked to incorporate (at minimum) a suit hatch, a view port, an equipment/rover hatch, and two combined communications-power-instrumentation penetrations within the habitat design and structure ("3D-Printed Habitat Challenge Rules," 2018). These elements were used to score submissions based on "Missing Element Information" and "Missing Category Information." Minimum ceiling heights and usage of square footage were demonstrated in vertical and horizontal sections taken from the 3D model, created in Autodesk Revit.

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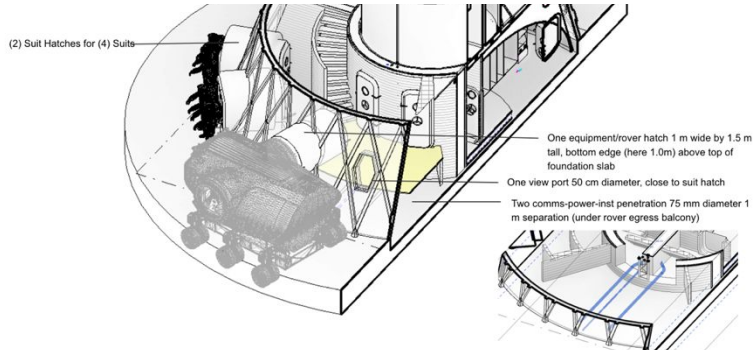

Figure 2. Required suit hatches, view port, equipment/rover hatch, and comms-power-inst penetrations within the Virtual Construction models

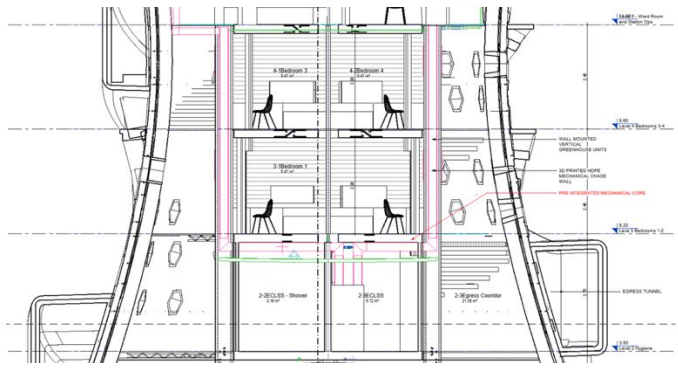

Figure 3. Required minimum ceiling heights demonstrated within BIM

The 100% Virtual Construction (BIM) Level 2 submission was required to have MEP and ECLSS design progressed to a Level of Development (LOD) 200 and the structural and pressure retaining walls and components progressed to LOD 400 ("3D-Printed…, 2018). The BIM Forum 2018 LOD Specification defines LOD 200 as a general model where elements are modeled with approximate quantities, size, shape, location, and orientation. LOD 400 defines model elements as specific assemblies, with complete fabrication and detailing information in order to derive full constructability from the model.

The competition asked teams to consider 3D BIM Habitat Modeling in order to "depict an accurate representation of physical conditions, printing environment, and assets" of the construction site, 4D BIM Construction Sequencing to "graphically represent both permanent and temporary facilities on site during multiple phases of the construction process," Equipment and Material Flow "to demonstrate the movements of construction methods including, but not limited to 3D-printers, material handling, and all autonomous systems on site," and finally a Virtual Mockup of the building model and construction process, or "the model used to design and analyze the construction of the facility components to improve their planning and constructability" ("3D-Printed Habitat... Competition Rules"). Scoring points were assigned based on the aforementioned criteria.

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Table 1. Virtual Construction Level 2 (100% Design) Scoring Rubric

Scoring Area	*Requirements*	*Points*
Completeness of 3D Model	a) Element Level of Development	125
	b) System Information	125
BIM Use Functionality	a) Layout/Efficiency	250
	b) Constructability/Robustness	250
Aesthetic Representation	Aesthetic Representation	250
4D Model Realism (Bonus)	a) Construction Sequencing	100
	b) Material & Equipment Flow	70
	c) Virtual Mockup	30
Total		1000 + 200 (bonus

Software Workflow. Autodesk BIM 360 was used as a cloud based platform for all project collaborators (architects, MEP consultants, structural consultants, and BIM specialists) to work within a unified Revit model updated in real-time. Certain features of BIM 360 were utilized to flag and revise clashing or non-integrated elements between disciplines. The softwares Rhino and Grasshopper (McNeel) were used for parametric modelling of the habitat's regolith structural shell. A workflow between Dynamo and Revit was considered for parametric design of the shell but given limited time to produce deliverables for the competition Rhino and Grasshopper proved to be a more expedient method. Once complete the structural shell was imported into Revit. The interior of the habitat was designed natively in Revit. At that point Revit families were assigned based on material usage within the habitat. Once complete, the Revit file was imported into Autodesk Navisworks. Within Navisworks, model elements were assigned to construction schedule activities in order to create a 4D construction sequence. Durations were assigned to deployment of various elements to graphically represent their actual install time. For 3D-printed components the duration was calculated based on the printer's anticipated trajectory speed of printing as well as the volume of the sub-assembly of the habitat. The sequence itself was subdivided based on the levels of the habitat (5 in total).

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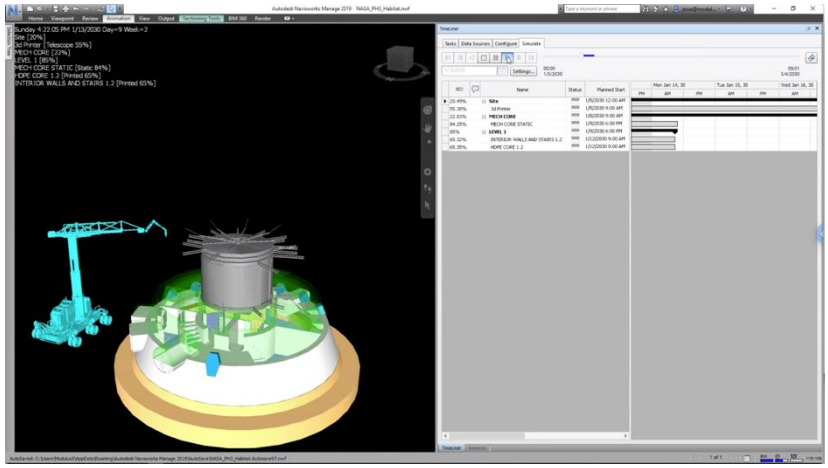

Figure 4. Still from 4D Construction Sequence.

Simulation of construction sequencing of the habitat (4D-BIM) demonstrated two mobile 3D-printers, as well as mobile platforms, coordinated in printing the habitat shell (composed of regolith concrete) and the interior's structure (composed of high-density polyethylene). Installation of habitat windows, a pre-integrated mechanical core, as well as pre-integrated hardware modules for laboratory experiments and communications were represented within the construction simulation (see Figure 4). In-progress components of the 3D-print were represented as "ghosted" green to better elucidate and capture those components of the model which were actively being additively manufactured over the course of the simulation.

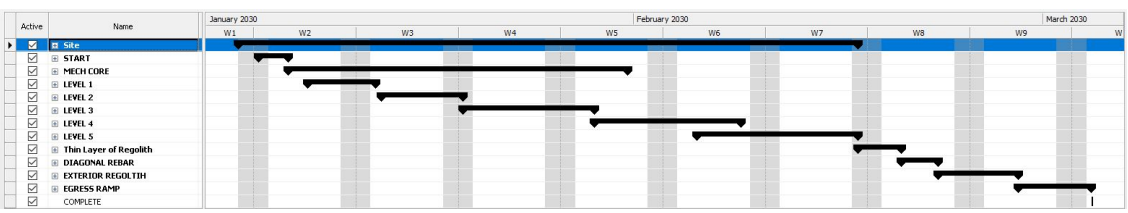

Figure 5. Construction Schedule developed in NavisWorks

Figure 6. Perspective view of the construction process and emplacement of a habitat window within the virtual design model.

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BIM & Sub-System Integration for Robotic Assembly. The complexity of MEP (mechanical, electrical, plumbing) installation tends to contribute to an understanding of buildings as one-off structures with little if any repeated assembly tasks and procedures. The potential to embed building utilities such as heating, electrical and plumbing components has been celebrated by early pioneers of additive construction such as Khoshnevis (Khoshnevis et al, 2005 and 2004), however standards for sub-system integration of MEP hardware are yet to be developed for integration with large-scale additively manufactured structures. For terrestrial AEC and future space habitat construction alike, building information modeling (BIM) represents a critical software capability to represent, plan, and manage all information relevant to a structure's construction elements and components within a single digital model.

Nevertheless, BIM alone is not sufficient for the direct planning of automated construction processes, both off-site and on-site. Construction tasks well suited to execution with industrial robotics include: installing thermal insulation, stacking, nailing elements together, or painting. Each of these tasks requires moderate force and is repetitive in nature. However a majority of building elements, particularly prefabricated components, are highly customized products. Even if a BIM provided information for assembly and installation of a component, there still would not be a clear path for automated or autonomous task execution by a robot. Additionally, building sites are dynamic; layout of materials, equipment paths, and weather are constantly changing in ways that cannot be represented in a static 3D model. BIM can be conceived more as a planning and management asset, and does not particularly benefit the use of robotics in construction (Meschini, Bock, et al, 2016). Merging BIM workflows with 4D construction sequencing geared towards robotic construction thus remains a critical task to advance autonomous construction at the building scale.

Dynamic BIM for Autonomous Systems. Communication delays and limited bandwidth over the course of a Mars mission will require human collaboration via teleoperation while autonomous surface site infrastructure are constructed. Humans will be able to provide high level strategic reasoning and problem-solving in the event of an error. Sensing, perception, and situational awareness will be critical capabilities for multiple robotic agents to collaboratively execute complex tasks within a construction sequence. Object recognition often requires fusing multiple sensing modalities, whereas a perception function can associate the sensed object with a reference that is understood in advance (Ambrose, Wilcox, 2010). Sensing approaches to date have combined machine vision, stereo vision, LIDAR, and RADAR. Perception approaches often start with CAD models or 3D-models created by a scan of the object in question. For autonomous systems to effectively operate on a building construction site, they must be able to refer to or derive a semantic model of the environment (Ferguson, "A-2D-3D..." 2019 and Ferguson, "Worksite..." 2019). It is extremely important that human-robot teams be able to access and contribute to a dynamically updated BIM (Correaa, 2016). Autonomous robotics for extra-terrestrial

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building construction will incorporate computer vision hardware and optical object-locating sensors such as: CCD cameras, 2D laserprofilers and 3D cameras (Salmi et al, 2016). In industry, 2D machine vision has been a standard solution to flexible object localization. In theory a vision system on a mobile robot could sense common worksite objects as well as events and a computer vision algorithm could not only detect objects but characterize them. At that point real-time building information modelling could enable building components to spawn, modify, or be revised in some way within a unified cloud-based model accessible by all agents.

A framework for all robotic and autonomous agents to reference a dynamic BIM model and perhaps even more importantly, the 4D construction sequence or schedule of the overall construction process, will be absolutely critical in the development and advancement of autonomous construction for both earth and space. Eventually, dynamic building information models could be used alongside extensive simulation and machine learning to anticipate errors and predict collisions, near-misses and safety issues in real-time (Ferguson, "A-2D-3D..." 2019 and Ferguson, "Worksite..." 2019). The framework for a dynamic BIM updated in real-time with the events, processes and milestones of the construction site itself will prove indispensable to human-machine collaboration within autonomous construction, as it will ensure a shared mental model of construction progress.

CONCLUSION

Wall apertures (windows), interior systems (partitions and flooring), structural reinforcement, building integrated sensor networks, among others will need to be robotically placed within 3D-printed structures at the appropriate time. Surface habitats will feature windows and apertures so that the crew may survey the surrounding terrain and maintain situational awareness at the site without conducting an EVA. Hard-shell modules that contain pre-integrated hardware such as ECLS or airlocks will most likely launch from Earth and assemble with additively manufactured structures. The deployment, positioning, and manipulation of hard-shell sub-systems will need to be autonomously executed by multiple mobile robots at the construction site. The timing and scheduling of emplaced habitat elements thus need to be closely correlated with the 3D-printing technology as well as the overall construction schedule for the habitat.

The 3D-printed structural prototypes and submissions by SEArch+ / Apis Cor to the Phase 3 3D-Printed Habitat Challenge demonstrate early advancements in autonomous construction relevant to future surface habitats. Once crewed missions arrive to Mars, site operations, field labour and regular repair and maintenance activities will be performed by both the crew and their robot counterparts collaboratively. In robotic precursor missions occurring prior to crew arrival, 3D-printing systems, machinery for ISRU regolith and water acquisition technologies, as well as space robotics for field labour, site excavation and preparation will need to coordinate and synchronize within a unified construction

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sequence, or 4D BIM plan. Remote supervision and control by ground support will occur with communications delays as well as limited bandwidth, further indicating the need for construction robotics to function and solve problems autonomously, while nonetheless consulting and conferring with mission control when help is needed. Testing and validation of pressurized 3D-printed structures remains a challenge still to be undertaken in large-scale earth-based additively manufactured prototypes anticipating the autonomous construction of habitats in future Mars mission scenarios.

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